ITAI 1378 • FINAL CAPSTONE

SmartFlow

Multi-Agent Traffic Monitoring and Zone Analytics

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Manual observation becomes a repeatable traffic report

The problem

Counting vehicles by hand is slow, inconsistent, and difficult to audit across multiple recordings.

The solution

SmartFlow detects, tracks, classifies traffic conditions, and saves evidence explaining every decision.

Three agents turn video into an explainable action

INPUT

Video or folder
validation + rotation

PERCEPTION

YOLO11 + ByteTrack
structured detections

ANALYSIS

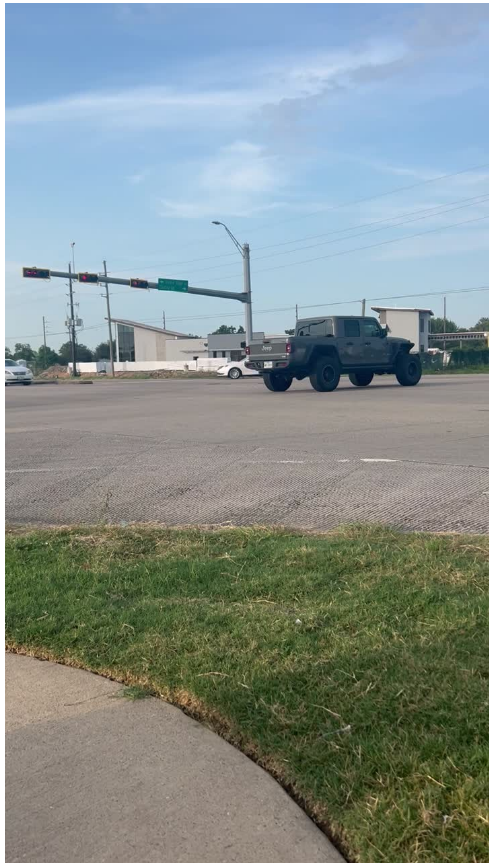
Counts + thresholds
LOW / MODERATE /
HIGH

REPORTING

Annotated video
report + CSV + JSON
trace

Dataclasses make every handoff inspectable and serializable.

Preprocessing solved a hidden orientation failure



BEFORE

MOV metadata ignored → 0 detections

AFTER

Clockwise rotation → vehicles detected

YOLO11 finds vehicles; ByteTrack preserves identity

Perception settings

• COCO vehicle classes
  car, motorcycle, bus, truck

• Confidence: 0.05

• Input size: 1280 px

• Every third frame

• Vertical center count line

Transparent thresholds make the decision easy to audit

HIGH	Peak ≥ 8 OR crossings/min ≥ 12	Create congestion alert
MODERATE	Peak ≥ 4 OR crossings/min ≥ 6	Save report; continue monitoring
LOW	Neither higher threshold reached	Save report; no alert

The report stores the measured values, matching threshold, and selected action.

Every run leaves evidence, not just console output

ANNOTATED VIDEO

Bounding boxes, IDs, labels, confidence, and count line

TEXT REPORT

Traffic level, class counts, reason, action, and latency

CSV + JSON TRACE

Frame observations and every agent handoff

Saved under results/ so the instructor can reproduce and inspect the run.

Ten scenarios test both normal operation and robustness

4

original videos

Busy, moderate, sparse,
and foreground occlusion

+

6

robustness variants

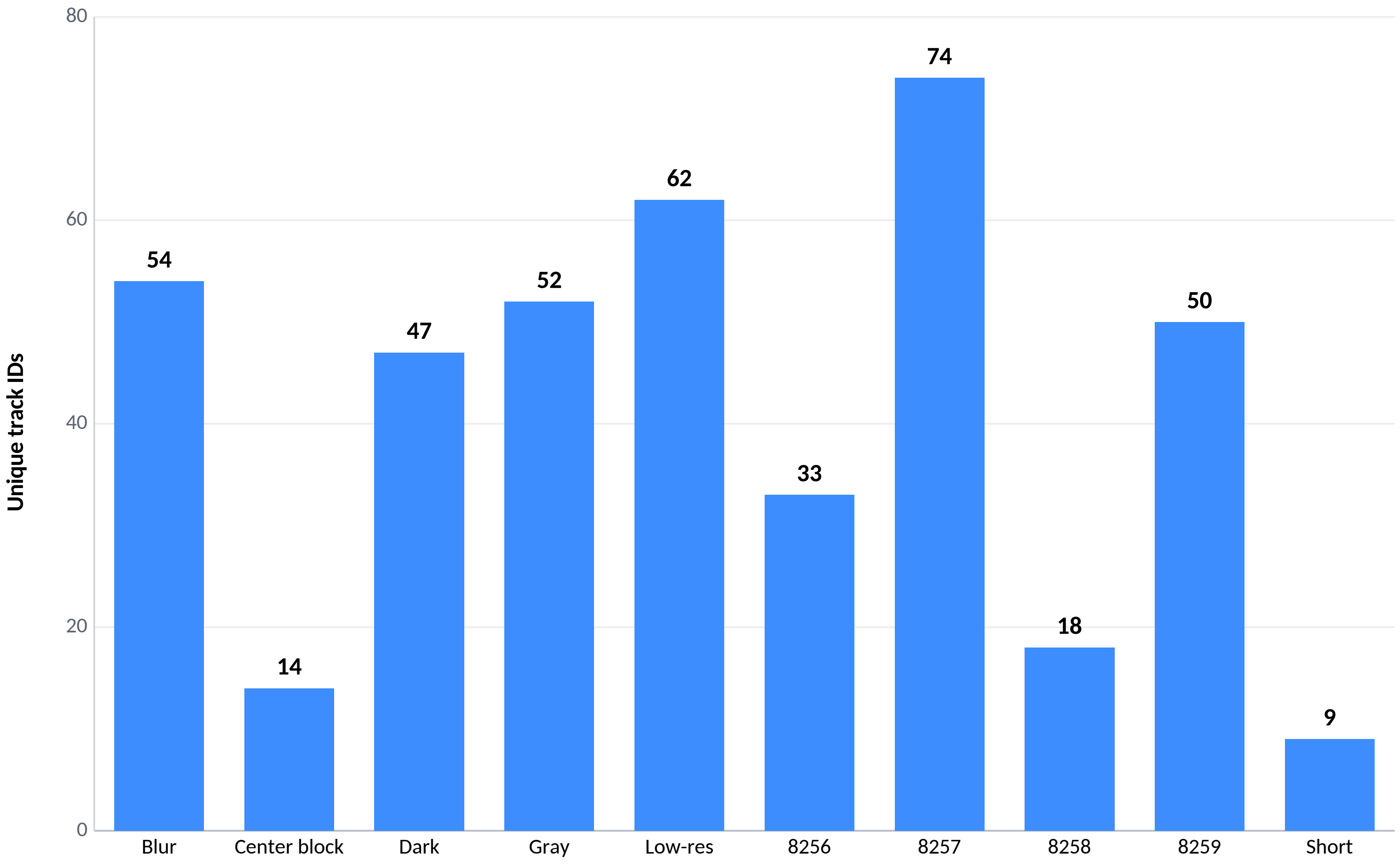
Blur • dark • grayscale
low resolution • center block
short clip

10 / 10

pipeline runs completed

Component metrics: tracks, crossings, peak vehicles, class counts, and speed

Traffic activity varied sharply across the test set



74

highest count IMG_8257

≈37

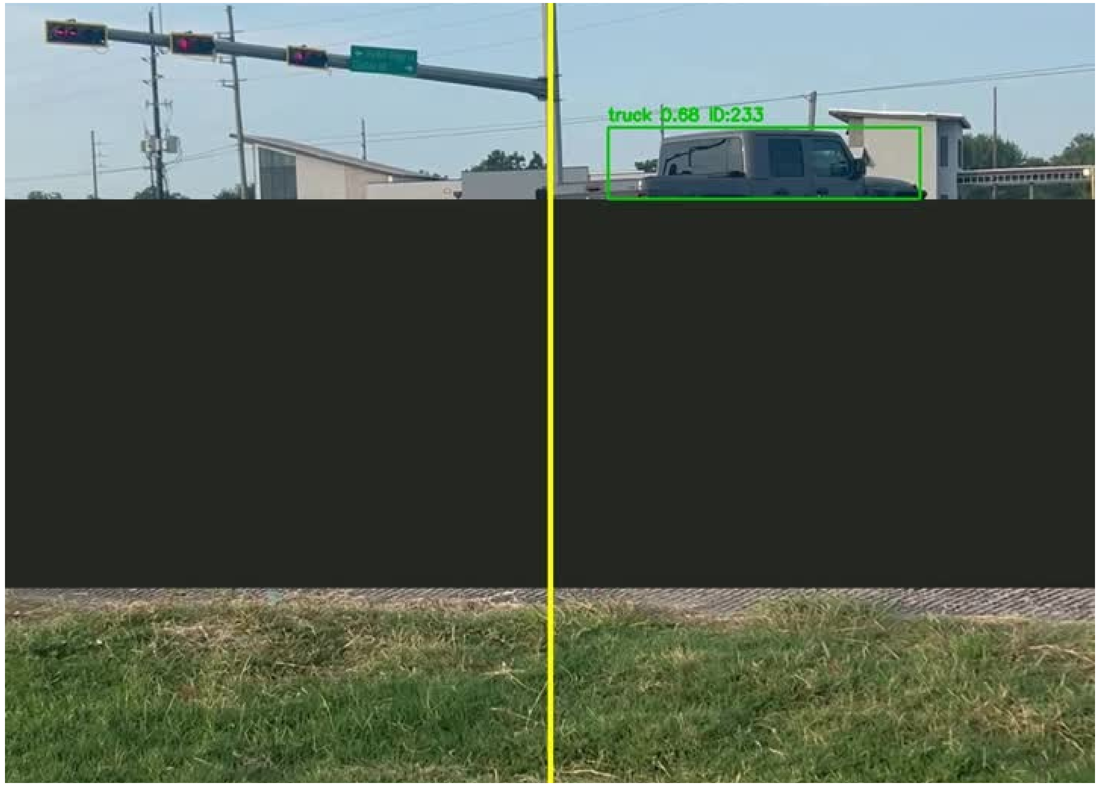
processed frames/sec mean on Colab T4

Original recordings produced usable traffic evidence

IMG_8257 • 74 tracks • 29 crossings • HIGH

IMG_8259 • 50 tracks • 20 crossings • HIGH

Failure cases reveal missed detections and ID fragmentation



OCCLUSION

50 → 14 tracks; 20 → 6 crossings
The block hides vehicles and breaks continuity.

LOW RESOLUTION

50 → 62 tracks despite the same scene
Pixelation creates new IDs and inflates the count.

SmartFlow completed the full agent pipeline

Perception → reasoning → action → saved trace

What worked

10/10 completed • explainable rules • real annotated evidence

Next improvements

Lane-specific lines • direction counts • local fine-tuning • privacy blurring

Repository includes code, documentation, sample data, results, and AI usage log.